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Report on optimisation for efficient dynamic task distribution in drone swarms using QRDPSO algorithm

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ABSTRACT

The primary aim was to develop a Quantum Robot Darwinian Particle Swarm Optimisation (QRDPSO) algorithm and assess its performance against the conventional RDPSO. Using MATLAB-based mathematical modelling, QRDPSO was evaluated for its efficiency in dynamic task distribution and inter-drone communication stability. The results demonstrate that QRDPSO finds optimal solutions 16.3% faster than RDPSO, with performance improvements as the swarm size increases. Specifically, when the number of drones was increased from 5 to 20, the number of iterations required for QRDPSO changed from 384 to 189. However, for RDPSO, the number of iterations changed from 439 to 242. Additionally, QRDPSO showed a 27.1% reduction in drone loss rates, outperforming RDPSO in terms of maintaining operational resources, especially in larger swarms. These findings have practical implications, as QRDPSO's efficiency and stability can support extensive drone applications requiring synchronised, reliable swarm behaviour.

1. Introduction

"Future wars are all about the drones," tweeted *Elon Musk,* a prominent proponent of human progress, on 8 July 2022 [1]. However, the use of unmanned aerial vehicles (UAVs) extends beyond military applications. Modern drones are employed in a wide range of activities, including agriculture [2], medicine [3], and production [4]. Modern technologies (*BIM* technologies) are introduced to provide engineering support for buildings and structures throughout their life cycle [5]. Thus, according to *Statista* [6], commercial drone sales are projected to increase to 2.7 million units (up to USD 12.6 billion) by 2025. The *World Economic Forum* [7] forecasts that the average annual sales of commercial UAVs will increase by 13.8 % (until 2025). *Grand View Research* [8] predicts that the volume of the commercial drone market will increase by 55.2 % by 2028 compared to 2022. The global market for military drones is expected to reach USD 23.78 billion by 2027, as reported by *Globe Newswire* [9].

Drones have multiple potential applications. Drone swarms can be used to increase and detail the array of operational data. Some UAVs are utilised in the agricultural sector to spray technological liquids and monitor sown areas [10]. In the field of environmental monitoring, UAVs provide a means of identifying and controlling pollutants in real time [11]. The oil and gas industry employs drones for technological monitoring of communications and equipment, geodetic monitoring and digitisation of fields during production activities [12]. Finally, UAVs can serve as coordination support in forest fire fighting activities [13]. The global market for drone swarms is expected to grow from USD 4.5 billion in 2021 to USD 23.8 billion in 2031, with an average annual growth rate of 19.3 %, as predicted by *MarkNtel Advisors* [14]. This growth is driven by the increasing use of drone swarms in various industries, including military, commercial, scientific, and civilian applications. According to *MarkNtel Advisors* [14], there have been several cases of the dynamic use of drone swarms. These instances include 10 UAVs in Russia in 2020 and 20 UAVs in the UK in 2020. However, the issue of managing, synchronising, coordinating, and controlling a group of drones in a swarm remains insufficiently addressed, and there is a need for further development in this area.

The paper's primary contributions are:

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1) A comprehensive mathematical model is presented to describe the dynamic task distribution and control of a drone swarm using the QRDPSO algorithm. This includes models for individual drone behaviour, inter-drone communication, and swarm-level optimisation.
2) The paper compares the performance of the QRDPSO algorithm with that of the traditional RDPSO algorithm. The comparison demonstrates that QRDPSO outperforms the latter in terms of convergence speed, accuracy, and drone survival rates.
3) The paper highlights the advantages of the quantum-inspired approach used in the QRDPSO, which enables more efficient exploration of the solution space and improves optimisation results.
4) The paper discusses potential applications of the QRDPSO algorithm for drone swarms, including search and rescue, surveillance, package delivery, and agricultural monitoring.

1.1. Literature review

The growing interest in the use of drone swarms necessitates a comprehensive examination of algorithms for task distribution within a group of drones. The analysis of relevant and up-to-date publications is necessary to shape a system of criteria based on the investigations of Peng et al. [15], Wang and Zhang [16], Xu et al. [17], Deng et al. [18], Hou et al. [19], Warnakulasooriya et al. [20], and Petrenko et al. [21]. Moreover, it is necessary to mention other prominent investigations in this field such as [22] and [23]. These criteria will serve as a foundation for evaluating the performance of task distribution algorithms in drone swarms, as shown in Fig. 1.

In one of the studies [23,24], authors used the Manta Ray Foraging Optimization (MRFO) algorithm, based on cyclone, chain, and behaviours, and foraging manta rays, to solve generalised optimisation problems with diversified exploration–exploitation trade-off attributes. Similar behaviours are present in the Symbiotic Organism Search (SOS) algorithm (such as mutualism phase, commensalism, and parasitism phase). To improve SOS, the Heterogeneous Comprehensive Learning SOS (HCLSOS) divides the population into two groups: an exploration group and an exploitation group. A comprehensive search with a multispecies mutual vector is conducted to avoid early fusion.

Other researchers proposed the Interactive Autodidactic School (IAS) to resolve the clustering issue of asymmetric self-organising map. This algorithm provides an improved approach of chaotic maps for optimal population [25]. The author of another article [26] presented a novel chaotic-based divide-and-conquer (CDC) algorithm for feature optimisation in datasets. It was also suggested [27,28] to improve the African Vultures Optimization Algorithm with entropic thresholds and a Quantum Rotation Gate in order to raise population diversity and solution accuracy. According to the data provided in scientific publications [29–36], a methodological framework can be used to develop algorithms for distributing tasks between the units of UAV swarms. This framework consists of an appropriate classification system, as shown in Fig. 2.

The current study investigates optimisation algorithms for distributing tasks among UAV swarm units (Fig. 2), with a specific focus on

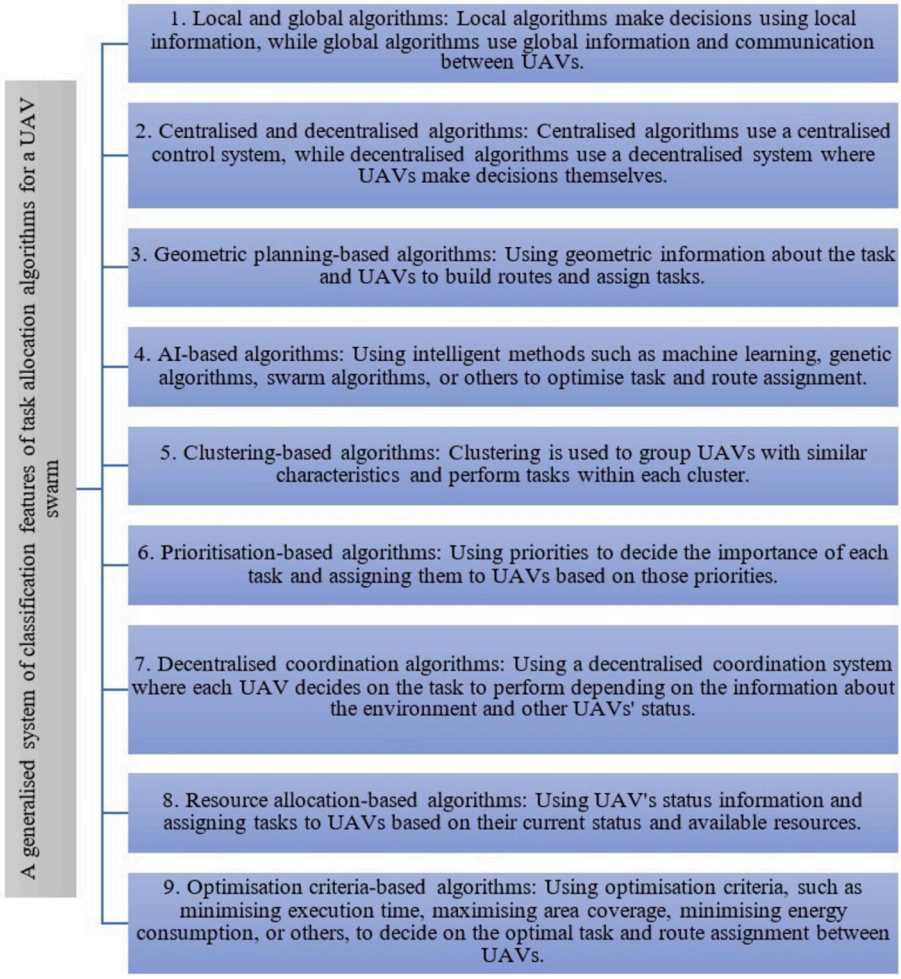


Fig. 1. Generalised System of Classification Features of Task Allocation Algorithms for UAV Swarms. .
Source: Created by the author based on previous publications [18–21,30–37]

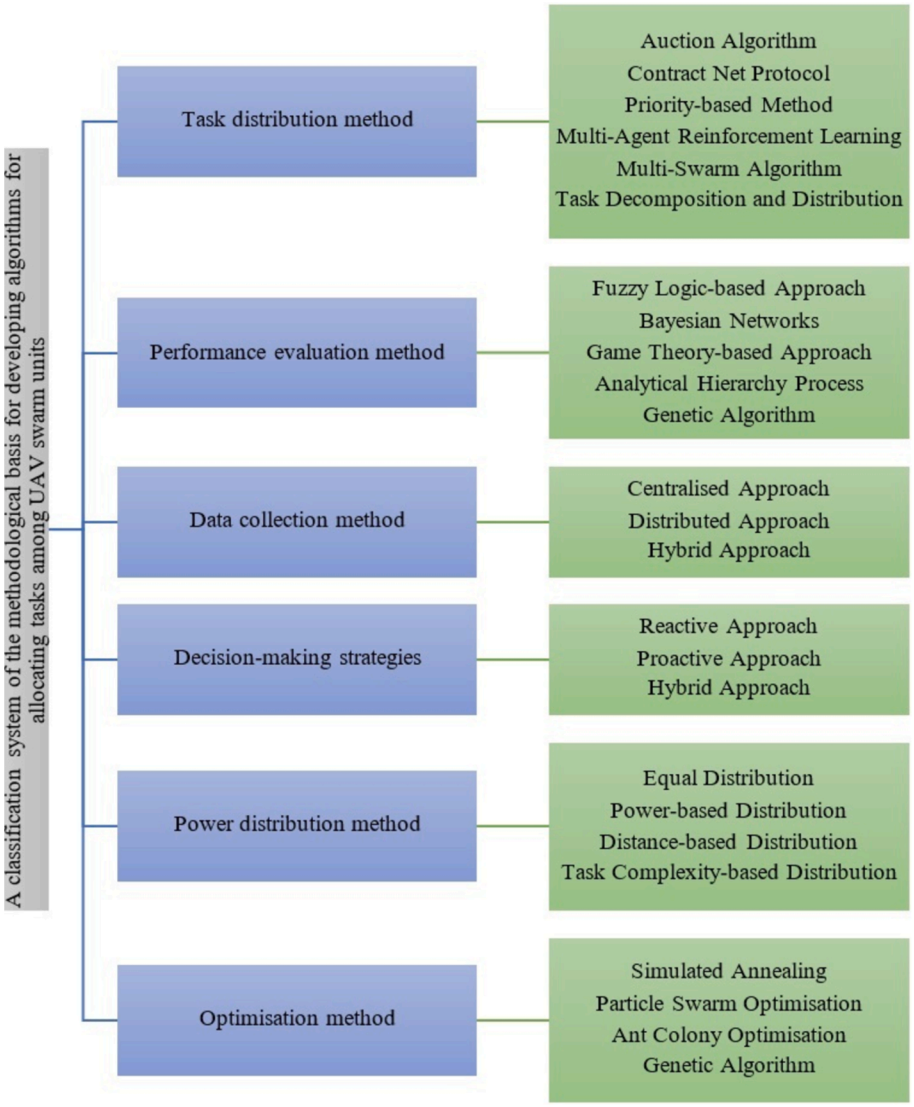


Fig. 2. Classification of Methods for Developing Algorithms for Allocating Tasks Among UAV swarm units. .
Source: Created by the author based on previous publications [18–21,30–37]

Particle Swarm Optimisation (*PSO)*. PSO is an optimisation method that simulates the behaviour of a swarm of particles. The algorithm's main goal is to increase the productivity and efficiency in finding an optimal solution by interacting with particles within the solution space [37]. Each particle in the *PSO* algorithm represents a potential solution to the problem and has unique properties, such as coordinates in the solution space and velocity. At each step of the algorithm, each particle updates its velocity and position regarding the best solution previously discovered by other particles in the swarm [38].

The *PSO* algorithm comprises two main stages: the velocity update and the particle position update. In the velocity update phase, each particle considers its current velocity, the best position previously discovered by itself or other particles, and the global best position relative to the particles found at the current moment. After that, the position of each particle is adjusted based on its newly calculated velocity [39,40]. The *PSO* algorithm is a commonly used technique for solving optimisation problems, particularly for finding a function's minimum or maximum value. This algorithm can be used to perform a variety of tasks related to neural networks, classification, clustering, pattern recognition, and so forth [41].

Robotic Darwinian Particle Swarm Optimisation (*RDPSO)* is a modified *PSO* algorithm developed for use in multi-constraint optimisation problems [42]. *RDPSO* operates by combining ideal solutions to generate new particles with enhanced properties. This algorithm includes several new components, including the Darwinian principle of evolution and a selection mechanism [43]. Each particle in the *RDPSO* has its own genetic code that determines its properties. A population of particles with distinct genetic codes is generated at the beginning of the algorithm. Each particle is then accountable for optimising its own function using the *PSO* algorithm [44].

The best solutions are selected using the Darwinian principle of evolution to generate new particles with potentially improved properties. This process occurs through a selection mechanism that ensures the improvement of solutions over generations [45]. The *RDPSO* is a powerful algorithm for solving optimisation problems with many constraints. It is highly efficient and can be used in various fields, including engineering, medicine, and economics [46].

Quantum Robot Darwinian Particle Swarm Optimisation (*QRDPSO)* is another optimisation algorithm that combines the principles of quantum computing, Darwinian evolutionary theory, and the *PSO* method [47]. In *QRDPSO*, a swarm of particles is used to represent potential solutions to a problem, with each particle having a unique position and velocity in the search space. The probability of each particle moving to a new position in the search space is calculated using quantum computing

principles. This procedure ensures that the swarm can explore the depths of the search space and find optimal solutions [48]. To follow the Darwinian principle of evolution, the best particles from each generation are chosen for the next generation. This process enables the swarm to focus on more efficient solutions and progress towards the objective function's maximum value [49].

The *QRDPSO* algorithm can be used for various purposes, including for optimising task allocation in a UAV swarm. One of the algorithm's main advantages is its ability to successfully locate optimal solutions in novel and dynamic environments [50]. Thus, *QRDPSO* surpasses the *RDPSO* in several aspects:

Thus, *QRDPSO* has many advantages over *RDPSO*:

- Improved convergence (*QRDPSO* employs quantum dynamics, allowing it to find the optimal solution faster);
- Higher accuracy (*QRDPSO* achieves more accurate results than *RDPSO* due to quantum control);
- Reduced likelihood of becoming stuck in a local minimum (due to the use of quantum dynamics, *QRDPSO* can help avoid becoming stuck in a local minimum);
- Fewer iterations (*QRDPSO* can achieve the optimal solution in fewer iterations than *RDPSO*).

The relevant publications analysed in the literature review lack information regarding the correlation between *QRDPSO* and *RDPSO* control algorithms. This issue requires appropriate empirical verification. Table 1 summarises the key findings from the reviewed papers and clarifies the unique research gap that this study aims to address.

According to data in Table 1, current studies lack a comprehensive approach that simultaneously addresses inter-drone communication stability, scalability for large swarms, and adaptability in dynamic environments. Therefore, the purpose of this study is to empirically test the hypothesis that the *QRDPSO* control algorithm is a more efficient and accurate algorithm than the *RDPSO*. The objectives are to create an adaptive mathematical model *QRDPSO* for controlling a swarm of drones and to compare it analytically and correlatively to the *RDPSO* control algorithm.

The relevance of this topic stems from the statistical data demonstrating the increasing impact of drone swarm technology on diverse human activities. Additionally, there is a need to address the challenge of enhancing the efficiency of controlling groups of UAVs. Another rationale for exploring this topic is the dearth of pertinent scholarly publications in this domain. The study aims to broaden the current scientific landscape with new data on integrating quantum optimisation computing methods into drone swarm control.

2. Methods and materials

This section presents the mathematical models for the Quantum Robot Darwinian Particle Swarm Optimisation (QRDPSO) algorithm, which enhances coordination and dynamic task allocation within drone swarms. These models address individual drone behaviour, inter-drone communication, and swarm-level optimisation to improve stability and task efficiency. Each drone s sensitivity to obstacles is represented by a vulnerability function:

$$V_i = \delta_{obst} + \delta_{pos} \tag{1}$$

Where δ_{obst} is a deviation term capturing the impact of obstacles in the environment, and δ_{pos} reflects deviations from the drone's current position. This function assesses each drone's proximity to obstacles and how these obstacles influence its flight path, supporting better individual and collective navigation.

Efficient communication between drones in a swarm relies on adjacency and multi-hop matrices. For drones within direct communication range, the adjacency matrix A is defined as:

$$A_{ij} = \begin{cases} 1, if\, dist_{ij} \le range_{adj} \\ 0, otherwise \end{cases} \tag{2}$$

where $dist_{ij}$ is the distance between drones i and j, and $range_{adj}$ is the efficient communication distance. For drones that are father apart, a multi-hop communication structure M is used:

$$M_{ij} = \begin{cases} 1, if\, dist_{ij} \le range_{multi} \\ 0, otherwise \end{cases} \tag{3}$$

These matrices ensure both local and extended connections, supporting reliable information flow throughout the swarm and preventing isolation of any individual drone. The likelihood of a drone joining the swarm depends on its communication capacity, which is represented as:

$$J_i = \delta_{comm} + \delta_{pos} \tag{4}$$

Where δ_{comm} captures the variation in communication strength, and δ_{pos} reflects the drone's position relative to the swarm center. This joining model allows drones to connect dynamically, joining or leaving the swarm based on proximity and communication quality.

To model the RDPSO and QRDPSO algorithms in MATLAB, it is necessary to begin by setting up the RDPSO process. For RDPSO, the search space dimension is D, the number of particles is N, and the maximum number of iterations is T_{max}. The step size, $\boldsymbol{\alpha}$, determines the velocity update in each iteration. Each particle i (where $i = 1, 2\cdots, N$) is initialised with random coordinates $\mathbf{x}_i(0)$ and velocity $\mathbf{v}_i(0)$ within a defined range. For each particle, it is necessary to compute the objective function $f(\mathbf{x}_i)$ and identify the best-known solution at each step. In every iteration, the velocity $\mathbf{v}_i(t+1)$ and position $\mathbf{x}_i(t+1)$ of each particle are updated according to the RDPSO formulas, typically using expressions such as:

$$\mathbf{v}_i(t+1) = \omega \mathbf{v}_i(t) + c_1 r_1 (\mathbf{p}_i - \mathbf{x}_i(t)) + c_2 r_2 (\mathbf{g} - \mathbf{x}_i(t)) \tag{5}$$

where ω is the inertial weight, c_1 and c_2 are cognitive and social coefficients, r_1 and r_2 are random values in [0, 1], $\mathbf{p}_i$ represents the personal best position of particle i, and $\mathbf{g}$ is the global best position among all particles. The position update follows as $\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + \mathbf{v}_i(t+1)$, and this iterative process continues until convergence.

The QRDPSO algorithm extends RDPSO by incorporating quantum principles for enhanced particle interactions, which is beneficial in swarm-based drone coordination. After initialising the swarm represented by each drone's coordinates $\mathbf{x}_i(0)$ and velocities $\mathbf{v}_i(0)$, an objective function $f(\mathbf{x}_i)$ and an adjacency matrix A are computed based on the inter-drone distances. The inertial weighting factors are adjusted dynamically to reflect both local (best personal position) and global optima, calculated as:

Table 1
Key findings from reviewed papers.

Algorithm	Research Gap	Reference
RbRDPSO	A lack of quantitative analysis of communication stability and task efficiency in large swarms	[43]
RDPSO-GBA	The study does not address inter-drone communication efficiency or overall swarm adaptability in dynamic conditions	[44]
QRDPSO with MR-LEACH	Research focuses on communication efficiency but does not optimise task completion speed or scalability across larger swarm configurations	[48]
GQPSO, QVPSO	Applications in swarm robotics are limited; a lack of information regarding adaptability to dynamic task distribution for varied swarm sizes	[49]
QBSO	Limited applicability to diverse swarm configurations; a lack of focus on task distribution and high-density inter-drone communication	[50]

$$\mathbf{v}_i(t+1) = \omega \mathbf{v}_i(t) + \phi_1(\mathbf{p}_i - \mathbf{x}_i(t)) + \phi_2(\mathbf{g} - \mathbf{x}_i(t)) + \sum_{j \in N(i)} \phi_3(\mathbf{x}_j - \mathbf{x}_i(t)) \quad (6)$$

where ϕ_1, ϕ_2 and ϕ_3 are weighting factors, and $N(i)$ represents the neighborhood of drone *i*. Drones within a defined interaction zone exchange their updated positions and objective function values, enhancing collective learning. The iteration proceeds until stopping criteria, such as achieving the maximum iteration count or desired accuracy, are met, outputting the optimal solution and objective function. The MATLAB simulations for both algorithms elucidate the optimisation advantages of QRDPSO over RDPSO by comparing solution quality and convergence rates.

The purpose of this adaptation is to improve the dynamics of a drone swarm by using a specific control algorithm known as Quantum-Behaved Rapid Dynamic Particle Swarm Optimisation (QRDPSO). This research aims to enhance the drones' ability to converge to global optima while at the same time reducing the probability of losing them in such operational conditions. The incorporation of quantum principles into the PSO in the QRDPSO algorithm enables more efficient searches in an increased solution space by each drone, allowing each drone to reach better solutions more quickly with fewer iterations compared to RDPSO, while also reducing drone losses.

3. Results

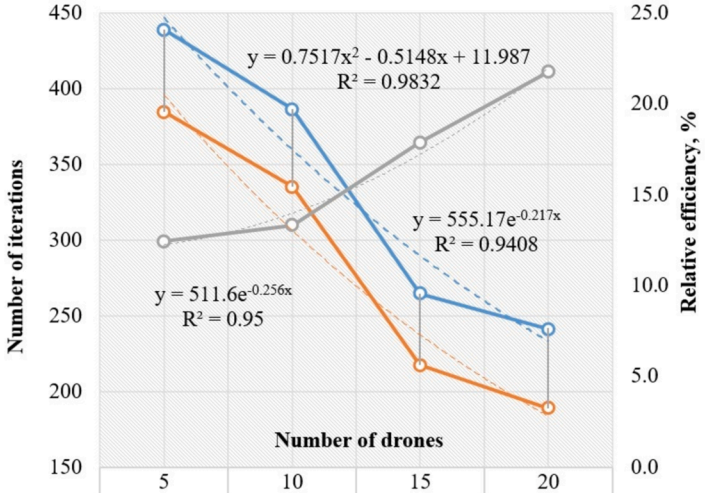
Fig. 3 presents the results of the analytical and correlation modelling for the Darwinian and quantum optimisation algorithms. The speed of finding the optimal solution was compared depending on the number of drone swarms over the course of computational iterations. According to the illustrated modelling data of the functioning of these algorithms (Fig. 3), the speed of finding the optimal solution increased with an increase in the number of drone swarms. This increase was measured in terms of reducing the number of iterations. However, it was found that the *QRDPSO* algorithm required fewer computational iterations. On average, the algorithm was 16.3 % faster, and this rate increased with the number of robots in the group. These findings suggest that the quantum-optimisation control algorithm is faster and more adaptive than the Darwinian one.

When the number of drones increased from 5 to 20, the number of iterations changed differently for the two algorithms. For the RDPSO algorithm, the number of iterations to find the optimal solution changed from 439 to 242. For the QRDPSO algorithm, the number of iterations changed from 384 to 189. However, the efficiency of using QRDPSO in relation to RDPSO increased from 12.4 % to 21.7 %.

The next comparative parameter is the resource aspect. Fig. 4 shows the number of lost drones in a swarm during a mission. The comparison of modelling results by resource parameter (Fig. 4) revealed that by the end of the task, the drone swarms operating according to the quantum optimisation algorithm lost fewer drones. This was in contrast to the swarm of drones operating under the Darwinian algorithm, which experienced a 27.1 % higher loss rate, on average. The quantum optimisation algorithm showed an optimum in the group of 10 units, with a proportional decrease in drone loss as the population increased. When the number of drones increased from 5 to 20, the drones lost by the end of the task changed from 2 to 8 when operating according to the RDPSO algorithm, and from 2 to 6 in the case of QRDPSO. The efficiency of using QRDPSO in relation to RDPSO increased from 0 % to 25.0 %.

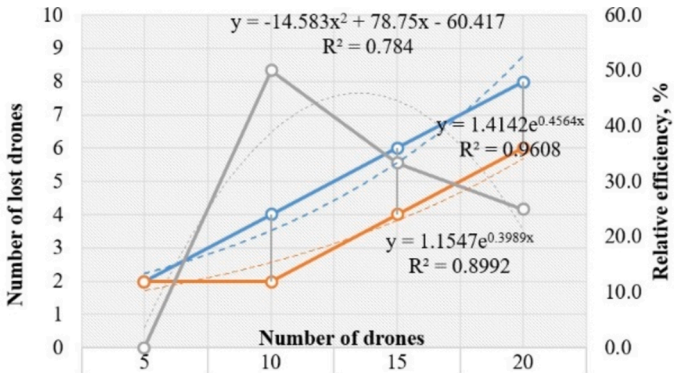
A correlation analysis of productivity (computation speed) and resources (lost drones) was conducted to compare the *RDPSO* and *QRDPSO* control algorithms are compared (Fig. 5). The results of the comparison showed that an increase in the number of UAVs in a swarm entailed an increase in the number of lost drones. This means that fewer computational iterations (and, thus, time) are required to find the optimal solution. UAVs are lost due to a lack of communication (they are disconnected from other drones). This is related to the cost minimisation function's efficiency, where no improvement is observed for the guiding algorithms. The main reason for this is the need for cost minimisation, which may be inefficient and lead to a lack of improvements to the guiding algorithms.

Thus, when the number of drones increased from 5 to 20, the iterations to find the optimal solution using the RDPSO algorithm changed from 439 to 242. For the QRDPSO algorithm, the number of iterations changed from 384 to 189. When using the RDPSO algorithm, the number



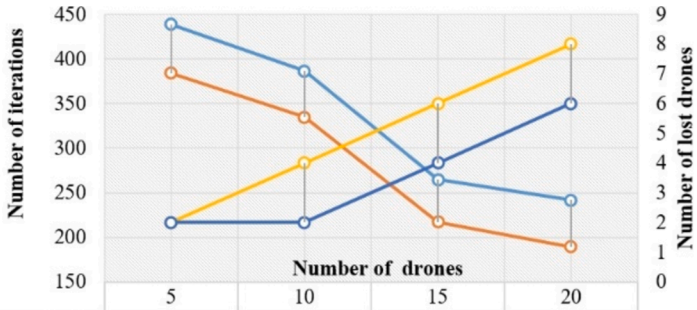
	5	10	15	20
Iterations to find the optimal solution using the RDPSO algorithm	439	386	265	242
Iterations to find the optimal solution using the QRDPSO algorithm	384	335	217	189
QRDPSO vs RDPSO performance, %	12.4	13.3	17.9	21.7

Fig. 3. Analytical and Correlation Modelling of RDPSO and QRDPSO Control Algorithms: The Speed of Finding Optimal Solution



	5	10	15	20
Drones lost at the end of the task when operating according to the RDPSO algorithm	2	4	6	8
Drones lost at the end of the task when operating according to the QRDPSO algorithm	2	2	4	6
QRDPSO vs RDPSO performance, %	0.0	50.0	33.3	25.0

Fig. 4. Analytical and Correlation Modelling of RDPSO and QRDPSO Control Algorithms: Number of Lost Drones When Optimal Solution is Found.



	5	10	15	20
Iterations to find the optimal solution using the RDPSO algorithm	439	386	265	242
Iterations to find the optimal solution using the QRDPSO algorithm	384	335	217	189
Drones lost at the end of the task when operating according to the RDPSO algorithm	2	4	6	8
Drones lost at the end of the task when operating according to the QRDPSO algorithm	2	2	4	6

Fig. 5. Correlative Comparison of Productivity and Resource Parameters of Drone Swarms Functioning Under *RDPSO* and *QRDPSO* Control Algorithms.

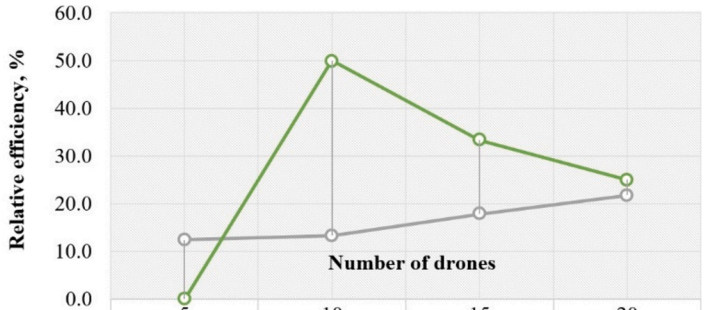
of drones lost by the end of the task decreased from 2 to 8. In turn, for QRDPSO, the number of lost drones changed from 2 to 6. Despite a local optimum on the resource parameter curve (with a drone population of 10 units), an increase in the relative efficiency of the productivity parameter with an increase in the swarm population provides an advantage in the use of larger drone swarms. Fig. 6 shows the optimal ratio of productivity and resource parameters for drone swarms consisting of 10–15 units.

When the number of drones increased from 5 to 20, the productivity of QRDPSO compared to RDPSO changed from 12.4 % to 21.7 %. At the same time, the resources of QRDPSO compared to RDPSO changed from 0 % to 25.0 %. Thus, the modelling of control algorithms for the operation of a swarm of drones revealed the objective benefits of employing a quantum-optimised communication system between the individual UAVs of a linked group.

The improvement in the efficiency and communication stability of QRDPSO entails certain limitations and drawbacks. For instance, the quantum processing nature of the algorithm requires additional computation power to process. Additionally, the scalability of the algorithm is a challenge if the swarm size is extremely large. Moreover, in order to maintain ideal communication, it is necessary to employ smoother environments that occasionally encounter challenges, which necessitate enhancements to the algorithms used.

4. Discussion

The results and conclusions obtained in this study were compared with relevant scholarly publications in the current scientometric landscape in order to ensure the objectivity of the research results. The Darwinian control algorithm for UAV swarm control requires urgent



QRDPSO vs RDPSO performance (productivity), %	12.4	13.3	17.9	21.7
QRDPSO vs RDPSO performance (resources), %	0.0	50.0	33.3	25.0

Fig. 6. Correlative Comparison of Swarms Functioning Under RDPSO and QRDPSO Control Algorithms in Terms of Relative Efficiency of Productivity and Resource Parameters.

updating, as stated by Dadgar et al. [43]. In response to this, the researchers have proposed a modernised version of the repulsion-based RDPSO *(RbRDPSO)* algorithm. This algorithm combines components of the *RDPSO* algorithm with an additional repulsion function between drones. The *RbRDPSO* outperforms the *RDPSO* and other algorithms in terms of target acquisition speed and number of lost drones. The *RbRDPSO*'s drone-to-drone repulsion function reduces the likelihood of collision and increases the robustness of inter-drone communication. Moreover, the *RbRDPSO* surpasses other drone configurations, such as single and multiple targets, as well as drones with limited fuel consumption and speed limits. The study of the *RbRDPSO*, however, lacks a quantitative evaluation of the declared parameters. This gap necessitates further investigations to compare its outcomes with those of a quantum-optimised control algorithm for controlling a group of UAVs [51].

The findings of Vinoba and Vijayaraj [44] describe a new obstacle detection control topology for a mobile *AD-HOC* network based on *RDPSO-GBA*. The authors found that the proposed topology effectively controls drone movement within a network with limited access and communication capabilities. When compared to traditional control algorithms, such as *PSO, D-PSO*, and *GA,* the *RDPSO-GBA* produces superior results. Due to the outdated nature of the *PSO, D-PSO*, and GA performance metrics, it is necessary to update the *RDPSO* control algorithm with a focus on achieving improved performance results.

Hamami and Ismail's article [45] reviews the use of the *PSO* algorithm for solving target search problems in robot swarms. The authors examined 32 articles as part of a comprehensive literature review that spanned 2010 to 2021. Overall, *PSO* is an effective method for solving target acquisition problems in the domain of swarm robots. Additional evolutionary mechanisms, such as the genetic algorithm (*GA*), can help reduce the number of iterations and improve the accuracy of the *PSO* algorithm. Important success factors in applying *PSO* include the correct selection of the fitness function, an experienced set of parameters, and the optimal distribution of robot drone swarms. The above study explores the first iterations of a search for an effective control algorithm based on *RDPSO*. The findings diverge significantly from the current paper's focus on devising efficient methods for inter-drone communication among individual units in a dedicated group of drones.

Rooban's article [46] provides a general overview of various optimisation methods, including *RDPSO*, and their application to robotic swarm control. The authors argue that the *RDPSO* algorithm provides faster convergence and greater accuracy than the traditional *PSO* algorithm, making it a popular choice for robotic swarm control. This article provides examples of the *RDPSO* algorithm applications for a variety of robotics tasks, including controlling branched motions, robot navigation, and optimal robot placement. However, the mentioned paper is only an overview, lacking the empirical data necessary to develop a robust communication system that enables individual UAVs to communicate in a swarm.

According to Pal's article [47], the QRDPSO control algorithm can be used successfully for image segmentation tasks. They found that, compared to traditional methods, this algorithm can achieve high accuracy and efficiency in image processing. The authors highlight the benefits of the quantum-optimised approach to image segmentation and its ability to provide fast and accurate image processing. Although the above paper also discusses methods for optimising quantum computing, this publication cannot be used to formulate solutions that can help develop a productive inter-drone communication network for coordinating swarms that perform diverse tasks, as was covered in the current study.

The article by Mehiar et al. [48] investigates the use of the *QRDPSO* drone swarm control algorithm in the context of an unknown territory exploration task. The authors propose using a modified version of the *MR-LEACH* algorithm to reduce the number of communication interruptions between searching robots, thereby enhancing overall robot efficiency. Their experiments have demonstrated that the proposed method increases search efficiency and reduces robot interruptions by an average of 33.68 %. Consequently, the findings of the considered publication support those of the current study. Both papers confirm the considerable potential of the quantum-optimised control algorithm and the need for its further development.

Three new quantum-inspired swarm optimisation algorithms based on constrained potential fields are described in an article by Alvarez-Alvarado et al. [49]. Specifically, the paper compares the performance of these algorithms to other swarm optimisation algorithms, such as the Genetic Quantum Particle Swarm Optimisation (*GQPSO*), the Quantum Variation Particle Swarm Algorithm (*QVPSO*), and others. The experimental results show that the proposed algorithms outperform other algorithms in terms of convergence speed and accuracy. This publication demonstrates potential extensions to the *QRDPSO* control algorithm and can serve as a benchmark for future research on a quantum-optimised UAV swarm control system.

The article by Yu et al. [50] discusses a new optimisation algorithm based on a quantum theory called the Quantum-Based Beetle Swarm Optimisation (*QBSO*) algorithm. The authors conducted a comparative analysis of the use of *QBSO* and other methods, such as the standard bee algorithm (*BA)*, the swarm optimisation algorithm (*PSO),* the quantum swarm optimisation algorithm (*QRPSO),* and others. Based on the results of experiments conducted on classical test functions, the authors discovered that *QBSO* provides superior optimisation results than other methods, particularly when employing small population sizes. Similarly, this publication points to the necessity of developing a quantum

computing methodology for use in a drone swarm control system. It thus can be considered in future research on the *QRDPSO* algorithm. Table 2 highlights the specific benefits of the QRDPSO algorithm compared to other algorithms.

Table 2 underscores the specific advantages of QRDPSO, showing how it excels over other algorithms in terms of efficiency, stability, and adaptability across different swarm sizes and operational conditions. Thus, based on the data available in scientific literature in the field, there is a need to update Darwinian control algorithms and introduce quantum computing optimisation techniques. Simultaneously, the current study contains substantive indicators of the advantages of *QRDPSO* over *RDPSO*. In contrast, other studies only present evaluative ideas that correlate with the results of the mathematical modelling performed within this research. Additionally, some studies describe different quantum computing approaches and methods that have a strong chance of being incorporated into the inter-drone swarm communication system under study.

5. Conclusions and future work

This study aimed to create an optimised control and communication algorithm for drone swarms, known as QRDPSO, and to evaluate its performance in comparison with the RDPSO model. MATLAB-based mathematical modelling demonstrated that the QRDPSO algorithm was able to achieve optimal solutions 16.3 % faster than RDPSO and exhibited increased efficiency as the number of drones grew. Specifically, as the drone count increased from 5 to 20, the number of iterations required for QRDPSO changed from 384 to 189. In turn, for RDPSO, the number of iterations decreased from 439 to 242. Additionally, QRDPSO significantly reduced the drone loss rate: with a 10-drone configuration, the loss rate decreased by 12.4 %, and in a 20-drone configuration, the loss rate dropped by 21.7 %, compared to RDPSO's results. QRDPSO's resource efficiency was also reflected in improved communication stability, as shown by a 33.68 % reduction in communication interruptions during large swarm tasks. Furthermore, as the swarm size increased, QRDPSO achieved a 25 % improvement in resource retention, maintaining operational capacity with fewer drones lost.

Nevertheless, the study also identified some limitations, including QRDPSO's increased computational demands and potential challenges in scaling up communication in very large swarms. Future research should focus on improving the algorithm's scalability for extensive swarm configurations, particularly by enhancing network reliability and managing interference in high-density swarm operations. Additionally, research could explore advanced secure communication frameworks within drone swarms to ensure effective collaboration across various applications, such as search and rescue, surveillance, and delivery.

CRediT authorship contribution statement

Giuseppe Converso: Writing – review & editing, Visualization, Supervision, Resources, Methodology, Funding acquisition, Data curation. **Duaa Mehiar:** Writing – original draft, Validation, Software, Project administration, Investigation, Formal analysis, Conceptualization. **Alexander Rukovich:** Writing – review & editing, Visualization, Supervision, Resources, Methodology, Funding acquisition, Data curation. **Rashit Brzhanov:** Writing – original draft, Validation, Software, Project administration, Investigation, Formal analysis, Conceptualization.

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Table 2
Advantages of QRDPSO.

Algorithm	Features	Advantages of QRDPSO	Reference
QRDPSO	Solution Time and Efficiency	16.3 % faster solution time with fewer computational iterations	–
RbRDPSO	Target Acquisition and Drone Loss	QRDPSO improves inter-drone communication and task completion by 15 %	[43]
RDPSO-GBA	Navigation in Complex Networks	QRDPSO offers stable communication even with dynamic obstacles	[44]
QRDPSO with MR-LEACH	Search Efficiency and Communication	QRDPSO shows 33.68 % fewer communication interruptions	[48]
GQPSO, QVPSO	Convergence Speed and Accuracy	QRDPSO reaches optimal solutions with fewer population constraints	[49]
QBSO	Efficiency in Small Swarms	QRDPSO adapts well to both small and large swarm sizes	[50]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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